# GAMA

Genomic Availability &
Metadata Analysis Tool

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# User Guide

## Overview

GAMA (Genomic Availability & Metadata Analysis Tool) is an R-based framework for efficiently surveying publicly accessible sequencing data for user-defined sets of species. It unifies NCBI Assembly, SRA, and BioSample query searches to generate availability summaries with a composite 'data richness' score, alongside ontology-based breakdowns of SRA modalities, GEO experiment linkage, and BioSample/BioProject-level replication skew. GAMA is intended to support feasibility assessments for *in silico* research on underutilised plants.

## Quick start

### Installation

GAMA is currently distributed via GitHub and can be installed using pak (Csárdi & Hester, 2025).

```
install.packages('pak')
pak::pak('JLewis-dev/GAMA')
```

### 1. Load package

```
library(GAMA)
```

### 2. Configure NCBI access to improve rate limits (recommended)

```
rentrez::set_entrez_key('YOUR_API_KEY')
```

### 3. Query NCBI databases using a list of species

```
RESULTS <- query_species(c('Vigna angularis', 'Vigna vexillata'))
```

### 4. Summarise data richness

```
SUMMARY <- summarise_availability(RESULTS)
print(SUMMARY)
```

### 5. Visualise data richness

```
plot_availability(SUMMARY)
```

### 6. Summarise SRA modality composition

```
META <- summarise_sra_availability(RESULTS)
print(META)
```

### 7. Visualise SRA modality composition

```
plot_sra_availability(META)
```

8. Extract filtered Assembly accession metadata

```
ASM <- extract_assembly_metadata(RESULTS, best = TRUE)
print(ASM)
```

9. Extract filtered SRA accession metadata

```
SRA <- extract_sra_metadata(RESULTS, species = 'Vigna vexillata', class = 'genomic')
print(SRA)
```

10. Cite

```
citation('GAMA')
```

---

## Dependencies

GAMA is implemented in R (≥4.2.0) and built on tidyverse tooling for data manipulation, iteration, and plotting (R Core Team, 2024; Wickham *et al*., 2023; Wickham *et al*., 2025; Wickham & Henry, 2026). NCBI Entrez eUtils queries are performed via rentrez (Winter, 2017; National Center for Biotechnology Information, 2026), and the SRA expxml field is parsed with xml2 (Wickham *et al*., 2026). Tabular outputs use tibble (Müller & Wickham, 2026), while plots are produced with ggplot2 (Wickham, 2016). Ensure that all dependencies are installed prior to running GAMA.

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## Functions

GAMA incorporates two modules:

Genomic Availability – Interrogates NCBI databases using rentrez to quantify per-species sequencing data availability. Searches use organism-restricted terms with retmax = 999999 and history tracking enabled. Search and summary requests are wrapped in retrying handlers with increasing back-off to improve robustness against transient NCBI failures. Accession summaries are retrieved in batches of 100 using entrez_summary(). When result counts exceed the directly returned ID set, GAMA uses stored Entrez history (web_history) to retrieve records in species-local batches. Internally, GAMA uses dplyr and tidyr for data manipulation, with purrr for functional iteration. Provenance (tool version, timestamp, databases, and search terms) is attached to all query and summary outputs.

### query_species()

- Input – character vector of binomial species names
- Output – named list of per-species query results for Assembly, SRA, and BioSample, including record IDs and accession counts

### summarise_availability()

- Input – list returned by query_species()
- Output – tibble (gdt_tbl) with one row per species containing Assembly, SRA, and BioSample accession counts, component scores (A, S, B), and a composite data richness score (score)

### plot_availability()

- Input – tibble returned by summarise_availability()
    - Optional arguments –
        - rank: character; species ordering method ('highest', 'lowest', 'A-Z', 'Z-A', 'input')
        - abbreviate: logical; abbreviate species names (TRUE, FALSE)
        - theme_fn: function; ggplot2 theme function (ggplot2::theme_minimal)
        - colours: named character vector; fill colours for Assembly, SRA, and BioSample segments (c(Assembly = '#E3F9F2', SRA = '#A8E6CF', BioSample = '#56C5A8'))
- Output – ggplot showing stacked bar charts of data richness components across species

Metadata Analysis – Retrieves SRA metadata by parsing the expxml field using xml2, extracting and normalising LIBRARY_STRATEGY values, and assigning ontology-based modality class and subclass using strict matching. When LIBRARY_STRATEGY is missing, uninformative, or explicitly 'other', a fallback rescue is applied using LIBRARY_SOURCE, LIBRARY_SELECTION, and TITLE fields. Experiments remain 'unknown' when no reliable signal is present, while assays that do not match any ontological category are retained as 'other'. GEO linkage is recorded by scanning experiment XML for GSE/GSM accessions, then cached with BioSample and BioProject identifiers to support downstream availability summaries and SRA skew analysis without re-parsing XML.

### summarise_sra_availability()

- Input – list returned by query_species()
    - Optional arguments –
        - species: character vector; species from list to include (NULL, one or more species names)
        - all: logical; include subclass-level columns (FALSE, TRUE)
        - include_geo: logical; append GEO summary columns (FALSE, TRUE)
- Output – tibble (gdt_tbl) with one row per species containing total SRA and class-level modality counts; caches UID-level profiles attached to the output tibble as attr(x, 'sra_profile') and profile metadata as attr(x, 'sra_profile_info') for downstream GEO overlay plotting and skew analysis

### plot_sra_availability()

- Input – tibble returned by summarise_sra_availability()
    - Optional arguments –
        - species: character vector; species from tibble to include (NULL, one or more species names)
        - rank: character; species ordering method ('highest', 'lowest', 'A-Z', 'Z-A', 'input')
        - abbreviate: logical; abbreviate species names (TRUE, FALSE)
        - theme_fn: function; ggplot2 theme function (ggplot2::theme_minimal)
        - colours: named character vector; class colours (c(genomic = '#E3F9F2', transcriptomic = '#A8E6CF', epigenomic = '#56C5A8', chromatin = '#2FA083', other = '#166A55', unknown = '#BDBDBD'))
- Output – ggplot showing proportional SRA modality composition across species

### plot_sra_geo()

- Input – tibble returned by summarise_sra_availability()
    - Optional arguments –

    - species: character vector; species from tibble to plot (NULL, one or more species names)
    - classes: character vector; modality classes to display (c('transcriptomic', 'epigenomic', 'chromatin', 'other', 'unknown'))
    - rank: character; species ordering method ('highest', 'lowest', 'A-Z', 'Z-A', 'input')
    - theme_fn: function; ggplot2 theme function (ggplot2::theme_minimal)
    - colours: named character vector; class colours (c(transcriptomic = '#A8E6CF', epigenomic = '#56C5A8', chromatin = '#2FA083', other = '#166A55', unknown = '#BDBDBD'))
    - alpha_vals: named numeric vector; transparency for GEO linkage (c('Not GEO-linked' = 0.25, 'GEO-linked' = 1))
- Output – ggplot for a single species, or a named list of ggplot objects for multiple species, showing per-modality GEO linkage using counts from cached profiles attached to the input tibble (attr(x, 'sra_profile'))

### extract_assembly_metadata()

- Input – list returned by query_species()
    - Optional arguments –
        - species: character vector; species from list to include (NULL, one or more species names)
        - best: logical; return the best assembly per species (FALSE, TRUE)
- Output – tibble (gdt_tbl) containing assembly-level metadata per retained assembly, including species, accession, level, n50, coverage, biosample, bioproject, submitter, release_date, and ftp_path

### extract_sra_metadata()

- Input – list returned by query_species()
    - Optional arguments –
        - species: character vector; species from list to include (NULL, one or more species names)
        - class: character vector; modality classes to filter by (NULL, one or more class names)
        - subclass: character vector; modality subclasses to filter by (NULL, one or more subclass names)
        - only_geo: logical; retain only GEO-linked experiments (FALSE, TRUE)
- Output – tibble (gdt_tbl) containing experiment-level SRA metadata, including entrez_uid, biosample, bioproject, strategy_raw, strategy_norm, class, subclass, geo_linked, gse_ids, and gsm_ids

### summarise_sra_skew()

- Input – tibble returned by summarise_sra_availability() with cached UID-level profiles attached to the input tibble (attr(x, 'sra_profile'))
    - Optional arguments –
        - species: character vector; species from tibble to include (NULL, one or more species names)
        - unit: character; skew unit ('bioproject', 'biosample')
        - class: character scalar; single modality class to analyse (NULL, one class name)
- Output – tibble (gdt_tbl) with one row per species containing unit count (BioProject or BioSample), boxplot summary statistics of experiments per unit (min, q25, med, q75, max), and Inverse Simpson index (eff)

### plot_sra_skew()

- Input – tibble returned by summarise_sra_skew()

- Optional arguments –
  - species: character vector; species from tibble to plot (NULL, one or more species names)
  - rank: character; species ordering method ('highest', 'lowest', 'A-Z', 'Z-A', 'input')
  - abbreviate: logical; abbreviate species labels (TRUE, FALSE)
  - show_points: logical; overlay per-unit points (TRUE, FALSE)
  - point_colour: character scalar; point colour ('#56C5A8')
  - theme_fn: function; ggplot2 theme function (ggplot2::theme_minimal)
  - colours: named character vector; box fill and line colours (c(box = '#E3F9F2', line = 'black'))
  - show_labels: logical; label each box with eff and n (TRUE, FALSE)
  - label_digits: integer; decimal places for eff labels (1L)
  - point_alpha: numeric; point transparency (0.25)
- Output – ggplot visualising replication skew across species on a log10 scale, with optional per-unit point overlays when the input skew tibble carries cached profiles (attr(x, 'sra_profile'))

---

# Metrics

## Data richness

*Score* = *A* + *S* + *B*, where *A*, *S*, and *B* are the transformed contributions of Assembly, SRA, and BioSample accession counts. *A* = *best* + ln(1 + *total* − *best*), with assemblies weighted as Complete = 10, Chromosome = 8, Scaffold = 5, and Contig = 2; *best* is the maximum-weight assembly, with ties broken by highest N50, and *total* is the sum of all accession weights. *S* = 2·ln(1 + *SRA*), and *B* = ln(1 + *BioSample*). This formulation prioritises high-quality assemblies while incorporating diminishing returns for highly sampled taxa.

## Ontology-driven classification

Subclasses and recognised term variants for experimental modality were derived by mining >220,000 *Arabidopsis thaliana* SRA accessions (last updated: 31 January 2026) before manual curation to capture common submitter variants and deprecated terminology (Tab. 1).

**Table 1.** Ontology for assigning SRA experiments to modality classes and subclasses based on normalised LIBRARY_STRATEGY terms.

| Class | Subclass | Recognised term variants (normalised) |
|---|---|---|
| Genomic | Amplicon-seq | amplicon; amplicon seq; amplicon sequencing; targeted amplicon; targeted amplicon sequencing; amplicon based sequencing; 16s amplicon; 16s sequencing; 16s rrna sequencing; 18s amplicon; its amplicon; its sequencing; metabarcoding |
| | Clone-based | clone; cloneend; poolclone; clone end; clone ends; clone end sequencing; clone based; clone based sequencing; bac end sequencing; bac end seq; fosmid end sequencing; fosmid end seq; cosmid end sequencing; cosmid end seq; pool clone; pool clone sequencing |

| | | |
|---|---|---|
| | RAD-seq | rad seq; rad sequencing; radseq; restriction site associated dna sequencing; restriction site associated sequencing; dd rad; dd radseq; double digest radseq; gbs; genotyping by sequencing; genotyping by seq |
| | Targeted-Capture | targeted capture; targeted sequencing; targeted seq; targeted resequencing; exome sequencing; exome seq; whole exome sequencing; wes; capture sequencing; capture seq; hybrid capture; target enrichment; panel sequencing; gene panel |
| | WGS | wgs; wga; wcs; wxs; finishing; whole genome sequencing; whole genome seq; whole genome shotgun; whole genome shotgun sequencing; whole genome resequencing; genome sequencing; genomic sequencing |
| Transcriptomic | Long-read | iso seq; isoseq; direct rna seq; direct rna sequencing; nanopore cdna; nanopore cdna seq; full length transcriptome |
| | RNA-seq | rna seq; ssrna seq; ribo seq; fl cdna; est; rnaseq; rna sequencing; transcriptome sequencing; transcriptome seq; transcriptome profiling; whole transcriptome sequencing; whole transcriptome seq; total rna seq; total rna sequencing; stranded rna seq; strand specific rna seq; poly a rna seq; polya rna seq; cdna sequencing; cdna seq; full length cdna; full length transcript sequencing; single cell rna seq; scrna seq; single nucleus rna seq; snrna seq; tag based rna seq; tag seq; digital gene expression; dge |
| | small-RNA | ncrna seq; mirna seq; small rna seq; small rna sequencing; smrna seq; smrna sequencing; microrna sequencing; microrna seq; mirna sequencing; non coding rna seq; noncoding rna seq; lncrna seq; lncrna sequencing; small noncoding rna seq; small non coding rna seq; sirna seq; pirna seq |
| Epigenomic | ATAC-seq | atac seq; atac sequencing; atacseq; assay for transposase accessible chromatin sequencing; assay for transposase accessible chromatin seq; single cell atac seq; scatac seq; single nucleus atac seq; snatac seq |
| | Bisulfite-seq | bisulfite seq; mbd seq; medip seq; mre seq; bisulfite sequencing; whole genome bisulfite sequencing; wgbs; wgbss; reduced representation bisulfite sequencing; rrbs; rrbss; methylation sequencing; methyl seq; methylation seq; methylome sequencing; mbd sequencing; medip sequencing; mre sequencing |
| | CUT&RUN | cut run; cutandrun; cut run sequencing; cutandrun sequencing |
| | CUT&Tag | cut tag; cutandtag; cut tag sequencing; cutandtag sequencing |

| | | |
|---|---|---|
| | ChIP-seq | chip seq; rip seq; chip sequencing; chipseq; chip dna seq; chip dna sequencing; chip exo; chip exo seq; chip exo sequencing; rna immunoprecipitation sequencing; rna immunoprecipitation seq; rip sequencing |
| | DNase-seq | dnase hypersensitivity; dnase seq; dnase sequencing; dnase i hypersensitivity; dnase i hypersensitivity sequencing; dnase i seq; dnase i sequencing |
| | FAIRE-seq | faire seq; faire sequencing; formaldehyde assisted isolation of regulatory elements sequencing; formaldehyde assisted isolation of regulatory elements seq |
| | MNase-seq | mnase seq; mnase sequencing; micrococcal nuclease sequencing; micrococcal nuclease seq; nucleosome mapping; nucleosome positioning sequencing |
| | SELEX | selex; selex seq; selex sequencing; ht selex; high throughput selex |
| Chromatin | 3C-based | 3c; 4c; 5c; capture c; promoter capture c; hichip; plac seq; plac-seq |
| | ChIA-PET | chia pet; chia pet seq; chia pet sequencing; chromatin interaction analysis by paired end tag sequencing; chromatin interaction analysis by paired end tag seq |
| | Hi-C | hi c; hic; hi c seq; hi c sequencing; chromosome conformation capture carbon copy |
| | TCC | tethered chromatin conformation capture; tcc; tcc seq; tcc sequencing; tethered chromatin conformation capture sequencing |
| Other | Other | custom; custom sequencing; custom protocol; metagenomic; metagenome sequencing; metagenomic sequencing; metatranscriptomic; metatranscriptome sequencing; metatranscriptomic sequencing; proteomic; proteomics; metabolomic; metabolomics; chip array; microarray; genotyping array; snp array; array; imaging; optical mapping; optical map; validation; pcr; qpcr; rt pcr; rt qpcr; library construction; sequence; sequencing; test; pilot; control; spike in; spike in control |

## Replication skew

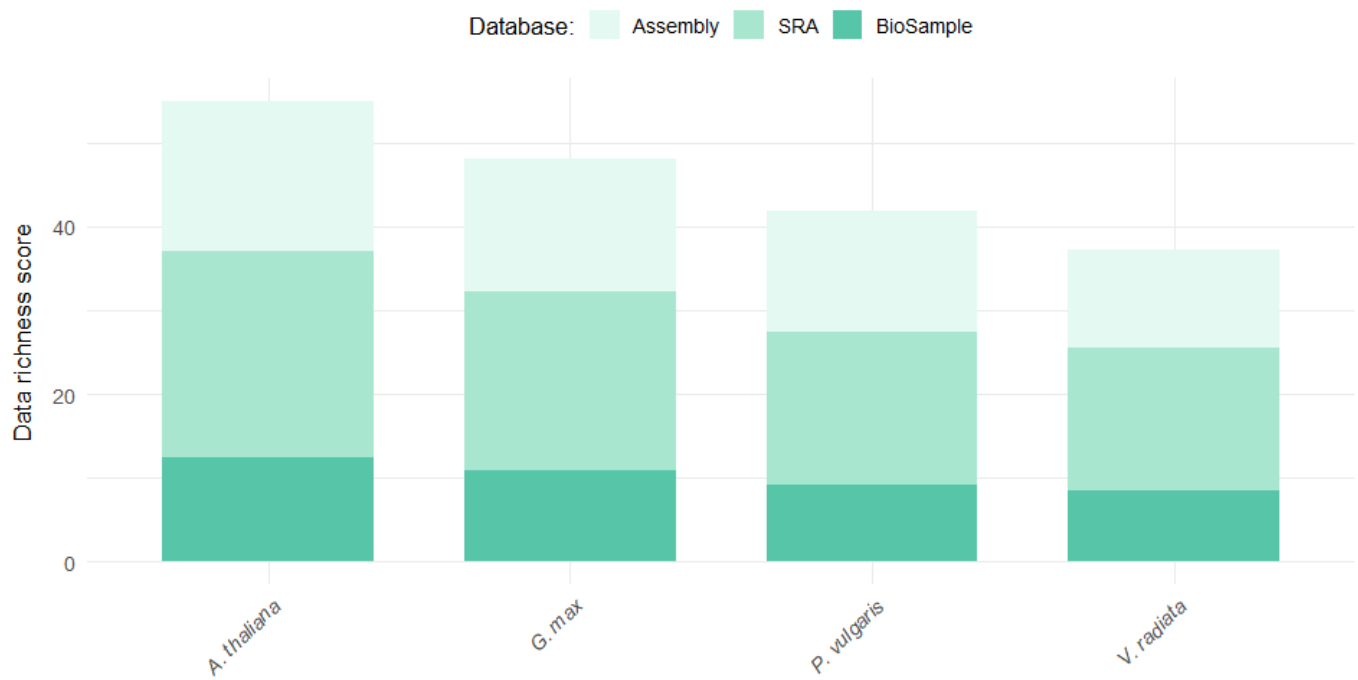
$eff = 1 / \Sigma(p_i^2)$, where $p_i = n_i / N$, $n_i$ is the number of experiments in unit $i$ (BioProject or BioSample), $N = \Sigma n_i$ is the total number of experiments, and $n$ is the number of distinct units. This is the Inverse Simpson index, expressed as an 'effective number' of units: $eff = 1$ when all experiments come from a single dominant BioProject/BioSample (Simpson, 1949), and $eff$ increases towards $n$ as experiments are distributed more evenly across many units (Hill, 1973). In SRA, experiments-per-unit are typically long-tailed (many low-count units with a small number of very

large projects/samples), so the squared-proportion weighting makes *eff* appropriately sensitive to dominance (replication skew) without being inflated by numerous singletons, providing a compact, comparable measure of independent replication alongside raw experiment counts (Jost, 2006).

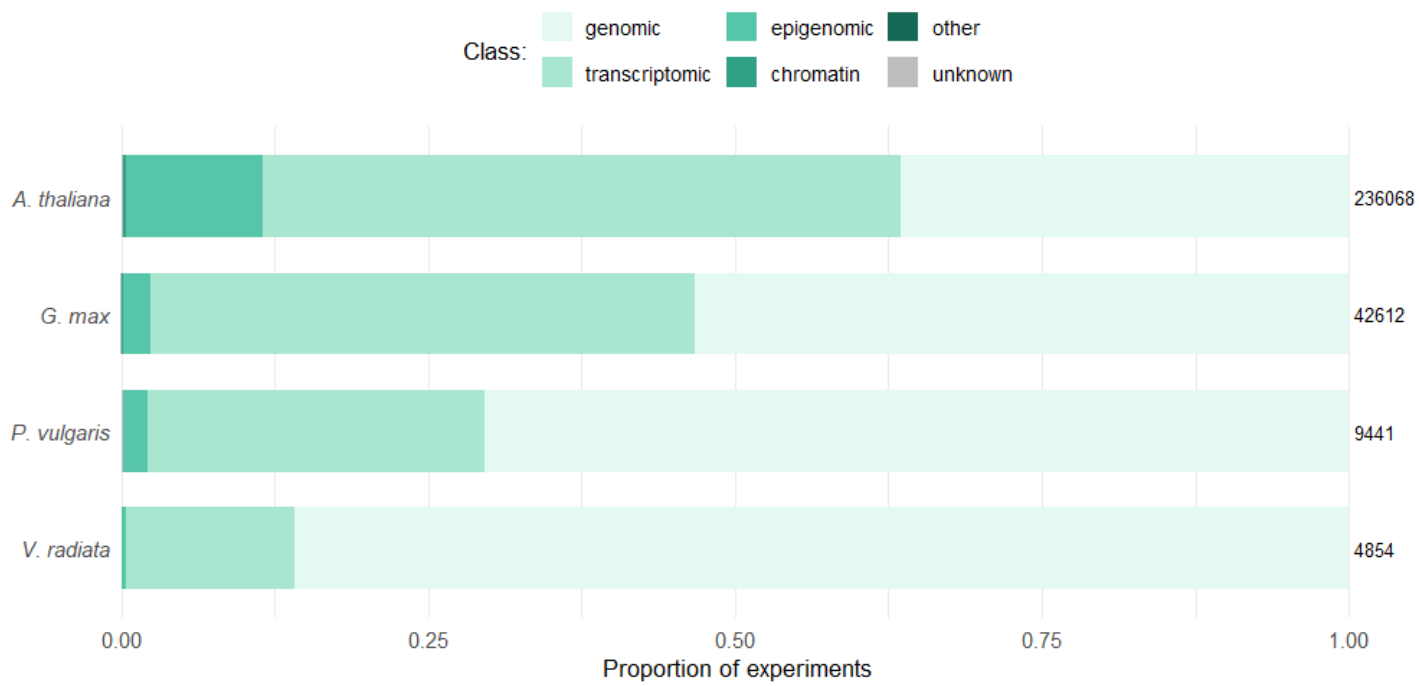
---

# Examples

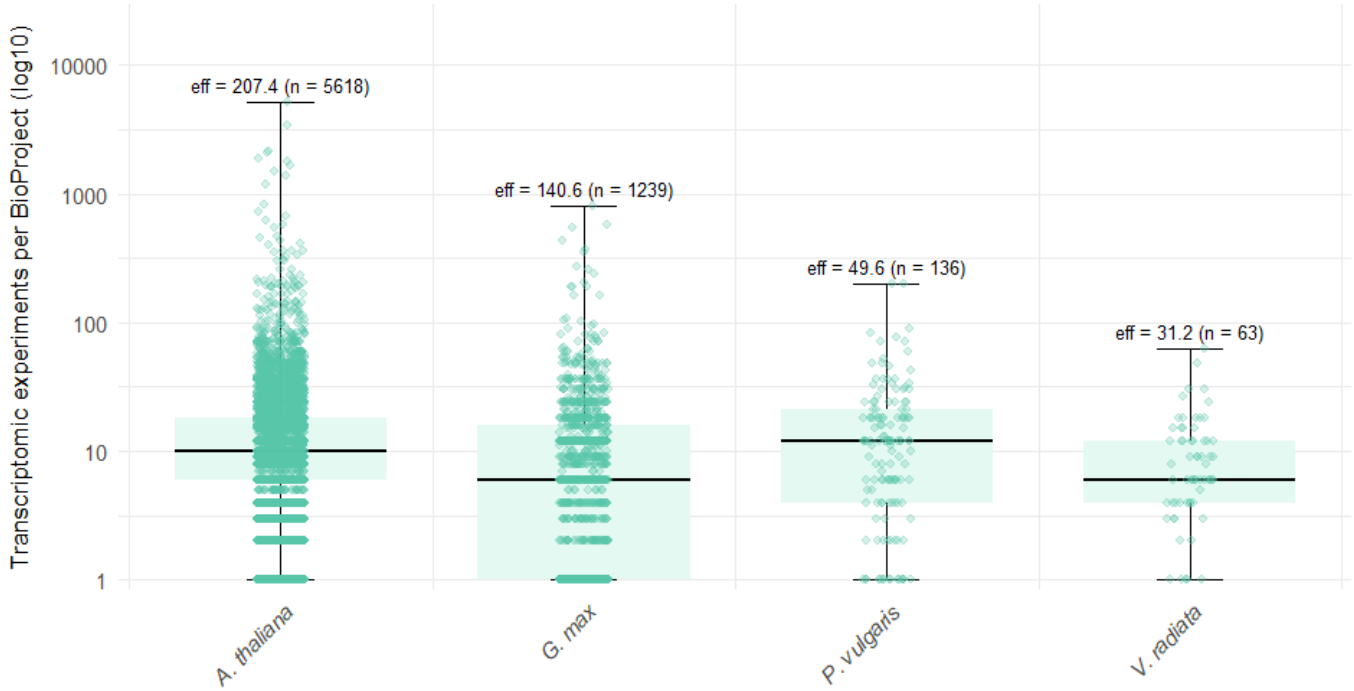
## Data richness plot




## SRA modality plot



## SRA skew plot




## Limitations

- Dependent on NCBI metadata quality
- Runtime increases with species list size
- Novel protocols may not be fully captured by the ontology
- Results should be interpreted cautiously during early development

## Development

GAMA aims to comprehensively summarise the 'metadata landscape' underlying public sequencing resources. Development is focussed on BioSample experimental parameters, extracting structured summaries of core descriptors (e.g. tissue, development stage, genotype, treatment, provenance). This is an order of magnitude more challenging than current availability counting because metadata fields are not enforced consistently across submitters and repositories. Heterogeneous free-text annotations, missingness, and study-specific terminology limit comparability, necessitating highly sophisticated normalisation and ontology-driven parsing.

# References

**Csárdi G. and Hester J. (2025)** pak: Another Approach to Package Installation. R package version 0.9.2.

**Hill M.O. (1973)** Diversity and evenness: a unifying notation and its consequences. *Ecology*, **54**: 427-432.

**Jost L. (2006)** Entropy and diversity. *Oikos*, **113**: 363-375.

**Müller K. and Wickham H. (2026)** tibble: Simple Data Frames. R package version 3.3.1.

**National Center for Biotechnology Information (2026)** Entrez Programming Utilities. National Library of Medicine. Available at: https://www.ncbi.nlm.nih.gov (Accessed: 31 January 2026).

**R Core Team (2024)** R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

**Simpson E.H. (1949)** Measurement of diversity. *Nature*, **163**: 688.

**Wickham H. (2016)** ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag, New York.

**Wickham H., François R., Henry L., Müller K. and Vaughan D. (2023)** dplyr: A Grammar of Data Manipulation. R package version 1.1.4.

**Wickham H., Vaughan D. and Girlich M. (2025)** tidyr: Tidy Messy Data. R package version 1.3.2.

**Wickham H. and Henry L. (2026)** purrr: Functional Programming Tools. R package version 1.2.1.

**Wickham H., Hester J. and Ooms J. (2026)** xml2: Parse XML. R package version 1.5.2.

**Winter D.J. (2017)** rentrez: An R package for the NCBI eUtils API. *The R Journal*, **9**: 520-526.